SAGAR PUBLIC SCHOOL

Integrated Programme • Topic: Work, Power & Energy

JEE MAIN Practice Sheet – 3

Name: _____ **Class / Sec:** _____ **Date:** _____

Time Allowed: 60 Mins • Total Questions: 20 • 12 from JEE Main PYQs + 8 from Module

A Energy & Conservation (Q1–8) **B** Work–Energy Theorem (Q9–14) **C** Power (Q15–17) **D** Vertical Circular Motion (Q18–20)

SECTION A – Energy & Application of Energy Conservation (Q1–Q8)

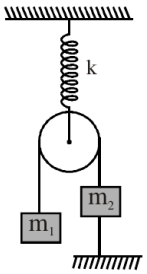
Q1. *[PYQ · JEE Main 2017, 8 Apr Morning]*

An object is dropped from a height h from the ground. Every time it hits the ground it loses 50% of its kinetic energy. The total distance covered as $t \to \infty$ is:

(A) $3h$ (B) ∞ (C) $\frac{5}{3}h$ (D) $\frac{8}{3}h$

Q2. *[Module · Ex. S-1, Q25]*

In the figure shown, pulley and spring are ideal. Find the potential energy stored in the spring ($m_1 > m_2$).

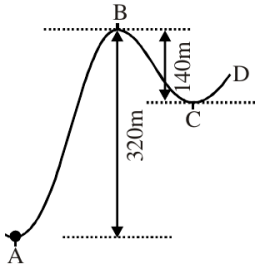


Q3. *[PYQ · JEE Main 2023, 6 Apr Evening]*

A body is dropped on ground from a height h_1 and after hitting the ground, it rebounds to a height h_2. If the ratio of velocities of the body just before and after hitting ground is 4, then percentage loss in kinetic energy of the body is $\frac{x}{4}$. The value of x is _____.

Q4. *[Module · Ex. O-1, Q18]*

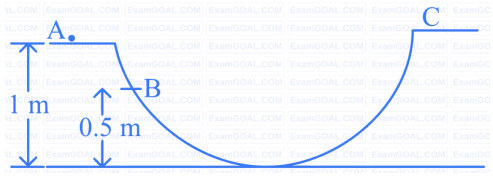
Track OABCD (as shown in figure) is smooth and fixed in vertical plane. What minimum speed has to be given to a particle lying at point A, so that it can reach point C?



(A) 60 m/s (B) 100 m/s (C) 70 m/s (D) 80 m/s

Q5. *[PYQ · JEE Main 2024, 30 Jan Morning]*

A particle is placed at the point A of a frictionless track ABC as shown in figure. It is gently pushed towards right. The speed of the particle when it reaches the point B is: (Take $g = 10\,\text{m/s}^2$).

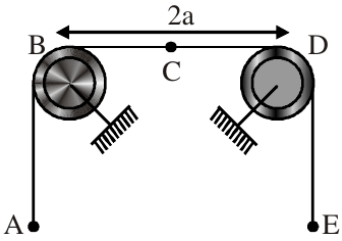


(A) $2\sqrt{10}$ m/s (B) 10 m/s

(C) $\sqrt{10}$ m/s (D) 20 m/s

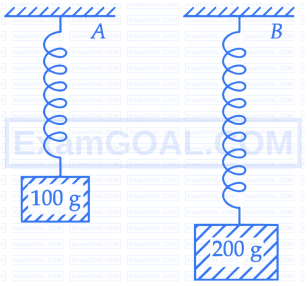
Q6. *[Module · Ex. S-2, Q7]*

A light string ABCDE whose mid point is C passes through smooth rings B and D, which are fixed in a horizontal plane distance $2a$ apart. To each of the points A, C and E is attached a mass m. Initially C is held at rest at O (mid point BD) and is then set free. What is the distance OC when C comes to instantaneous rest?



Q7. *[PYQ · JEE Main 2026, 23 Jan Morning]*

Two blocks with masses 100 g and 200 g are attached to the ends of springs A and B as shown in figure. The energy stored in A is E. The energy stored in B, when spring constants k_A, k_B of A and B, respectively satisfy the relation $4k_A = 3k_B$, is:



(A) $4E$ (B) $2E$ (C) $\frac{4}{3}E$ (D) $3E$

Q8. *[PYQ · JEE Main 2022, 26 Jul Morning]*

As per the given figure, two blocks each of mass 250 g are connected to a spring of spring constant $2\,\text{N m}^{-1}$. If both are given velocity v in opposite directions, then maximum elongation of the spring is:



(A) $\frac{v}{2\sqrt{2}}$ (B) $\frac{v}{2}$ (C) $\frac{v}{4}$ (D) $\frac{v}{\sqrt{2}}$

SECTION B – Work–Energy Theorem (Q9–Q14)

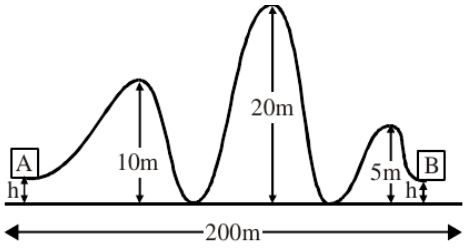
Q9. *[PYQ · JEE Main 2026, 6 Apr Morning]*

The rain drop of mass 1 g, starts with zero velocity from a height of 1 km. It hits the ground with a speed of 5 m/s. The work done by the unknown resistive force is _____ J. (take $g = 10\,\text{m/s}^2$)

(A) −8.75 (B) −8.35 (C) −9.55 (D) −9.98

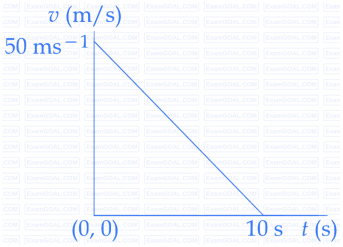
Q10. *[Module · Ex. S-1, Q13]*

A point object of mass 2 kg is moved from point A to point B very slowly on a curved path by applying a tangential force on a curved path as shown in figure. Then find the work done by external force in moving the body. Given that $\mu_s = 0.3$, $\mu_k = 0.1$. $[g = 10\,\text{m/s}^2]$



Q11. *[PYQ · JEE Main 2016, 10 Apr Morning]*

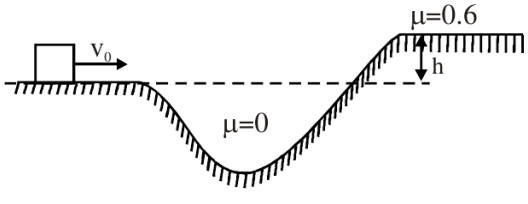
Velocity-time graph for a body of mass 10 kg is shown in figure. Work-done on the body in first two seconds of the motion is:



(A) 12000 J (B) −12000 J
(C) −4500 J (D) −9300 J

Q12. *[Module · Ex. O-1, Q10]*

In the figure, a block slides along a track from one level to a higher level, by moving through an intermediate valley. The track is frictionless until the block reaches the higher level. There a frictional force stops the block in a distance d. The block's initial speed v_0 is 6 m/s, the height difference h is 1.1 m and the coefficient of kinetic friction μ is 0.6. The value of d is



(A) 1.17 m (B) 1.71 m (C) 7.11 m (D) 11.7 m

Q13. *[PYQ · JEE Main 2024, 9 Apr Evening]*

A force $(3x^2 + 2x - 5)$ N displaces a body from $x = 2$ m to $x = 4$ m. Work done by this force is _____ J.

Q14. *[PYQ · JEE Main 2019, 9 Jan Evening]*

A force acts on a 2 kg object so that its position is given as a function of time as $x = 3t^2 + 5$. What is the work done by this force in first 5 seconds?

(A) 850 J (B) 950 J (C) 875 J (D) 900 J

SECTION C – Power (Q15–Q17)

Q15. *[PYQ · JEE Main 2023, 13 Apr Morning]*

The ratio of powers of two motors is $\frac{3\sqrt{x}}{\sqrt{x}+1}$, that are capable of raising 300 kg water in 5 minutes and 50 kg water in 2 minutes respectively from a well of 100 m deep. The value of x will be

(A) 16 (B) 4 (C) 2 (D) 2.4

Q16. *[Module · Ex. S-2, Q11]*

A car's engine can deliver 90 kW of power. The car's mass is 1000 kg. Assume the total resistive force is proportional to the velocity: $|F_{\text{friction}}| = \alpha v$. The drag coefficient α is $\alpha = 100\,\text{Ns/m}$. Car can maintain on a level road a maximum speed of _____ m/s?

Q17. *[PYQ · JEE Main 2025, 29 Jan Evening]*

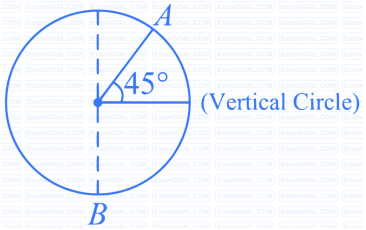
A sand dropper drops sand of mass $m(t)$ on a conveyer belt at a rate proportional to the square root of speed (v) of the belt, i.e., $\frac{dm}{dt} \propto \sqrt{v}$. If P is the power delivered to run the belt at constant speed then which of the following relationship is true?

(A) $P \propto \sqrt{v}$ (B) $P \propto v$
(C) $P^2 \propto v^3$ (D) $P^2 \propto v^5$

SECTION D – Vertical Circular Motion (Q18–Q20)

Q18. *[PYQ · JEE Main 2024, 4 Apr Evening]*

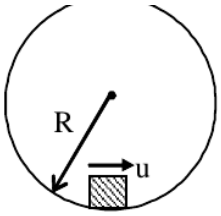
A body of m kg slides from rest along the curve of vertical circle from point A to B in frictionless path. The velocity of the body at B is: (given, $R = 14$ m, $g = 10\,\text{m/s}^2$ and $\sqrt{2} = 1.4$)



(A) 10.6 m/s (B) 19.8 m/s
(C) 16.7 m/s (D) 21.9 m/s

Q19. *[Module · Ex. O-1, Q29]*

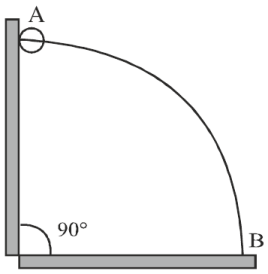
A particle is given an initial speed u inside a smooth spherical shell of radius $R = 1\,\text{m}$ that it is just able to complete the circle. Acceleration of the particle when its velocity is vertical is –



(A) $g\sqrt{10}$ (B) g (C) $g\sqrt{2}$ (D) $3g$

Q20. *[Module · Ex. JA, Q12 (JEE Advanced 2014)]*

A wire, which passes through the hole in a small bead, is bent in the form of quarter of a circle. The wire is fixed vertically on ground as shown in the figure. The bead is released from near the top of the wire and it slides along the wire without friction. As the bead moves from A to B, the force it applies on the wire is:



(A) Always radially outwards
(B) Always radially inwards
(C) Radially outwards initially and radially inwards later
(D) Radially inwards initially and radially outwards later

SELF ASSESSMENT (to be filled by the student after the test)

Time Taken: _____ min **Attempted:** _____ / 20 **Unattempted:** _____

Correct: _____ **Incorrect:** _____ **Accuracy:** _____ %

Section	Questions	Correct	Incorrect	Topics / concepts I must revise
A – Energy & Conservation	8			
B – Work–Energy Theorem	6			
C – Power	3			
D – Vertical Circular Motion	3			

Personal Feedback (what went well / what went wrong / silly mistakes / plan for next sheet):

_____

_____

_____

_____

Student's Signature: _____ **Teacher's Remark / Signature:** _____